Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VIII

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Sorting Algorithms:

Write programs to implement and compare the following sorting algorithms:

a. Bubble sort

Code:

```
def bubbleSort(array):

  for i in range(len(array)):
    for j in range(0, len(array) - i - 1):
      if array[j] > array[j + 1]:
        temp = array[j]
        array[j] = array[j+1]
        array[j+1] = temp

data = [-2, 45, 0, 11, -9]

bubbleSort(data)

print('Sorted Array in Ascending Order:')
print(data)
```

Output:

```
Sorted Array in Ascending Order:
[-9, -2, 0, 11, 45]
```

b. Insertion sort

Code:

```python
def insertionSort(array):

    for step in range(1, len(array)):
        key = array[step]
        j = step - 1

        while j >= 0 and key < array[j]:
            array[j + 1] = array[j]
            j = j - 1

        array[j + 1] = key


data = [9, 5, 1, 4, 3]
insertionSort(data)
print('Sorted Array in Ascending Order:')
print(data)
print("Divyamshu Singh S039")
```

Output:

```
.py
Sorted Array in Ascending Order:
[1, 3, 4, 5, 9]
Divyamshu Singh S039
```

c. Selection sort

Code:

```python
def selectionSort(array, size):

    for step in range(size):
        min_idx = step

        for i in range(step + 1, size):

            if array[i] < array[min_idx]:
                min_idx = i

        (array[step], array[min_idx]) = (array[min_idx], array[step])


data = [-2, 45, 0, 11, -9]
size = len(data)
selectionSort(data, size)
print('Sorted Array in Ascending Order:')
print(data)
print("Divyamshu Singh S039")
```

Output:

```
.py
Sorted Array in Ascending Order:
[-9, -2, 0, 11, 45]
Divyamshu Singh S039
```

Curiosity Questions:

Programming Questions

1. Write a Python program to sort a list of numbers in descending order using **Bubble Sort**.
2. Write a Python program to sort a list of numbers in descending order using **Insertion Sort**.
3. Write a Python program to sort a list of numbers in descending order using **Selection Sort**.

Theory Questions

4. What is sorting? Why is it required?
5. What is the basic idea of Bubble Sort?
6. How does Insertion Sort work?
7. How does Selection Sort work?
8. What is a **pass** in a sorting algorithm?
9. What is the difference between Bubble Sort and Selection Sort?

10. Which sorting algorithm is suitable when the data is already nearly sorted? Why?